

COURSE SYLLABUS

CS202 – PROGRAMMING SYSTEMS

1. GENERAL INFORMATION

Course name:	PROGRAMMING SYSTEMS
Course name (in Vietnamese):	Các hệ thống lập trình
Course ID:	
Knowledge block:	
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	CS162
Instructors:	Dinh Ba Tien

2. COURSE DESCRIPTION

The course is designed to provide students the object-oriented concepts, problem analysis, design and solving. The course helps students to differentiate between procedural programming and object-oriented programming. Students are also introduced and instructed with more advanced C++ concepts and features such as class, inheritance, polymorphism, operator overloading, templates, exception handling, and some design patterns. In particular, students could describe key object-oriented concepts, such as encapsulation, data hiding, inheritance and dynamic binding, and are able to solve problems using object-oriented programming. In addition, students are also guided to work in groups on a substantial computer program during the term, providing a user's manual and external design documentation. Students are also gained reading comprehension skills and learn to write simple reports in English.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Work independently or in groups to solve problems using object-oriented concepts.	LO9
G2	Perform reading comprehension skills, write and present simple reports in English	LO10
G3	Use object-oriented, analysis and design concepts to solve real-world problems.	LO1, LO3
G4	Comprehend the OOP concepts: class, object, data hiding, encapsulation, inheritance, dynamic binding...	LO1
G5	Describe the key differences between C++, C#, and Java on some OOP topics	LO1
G6	Solve problems in C++ using OOP	LO3, LO4, LO6

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Demonstrate the ability of working independently	TU
G1.2	Demonstrate working in pair or group on the project	TU
G2.1	Summarize the key features of text chapters.	TU
G2.2	Demonstrate an understanding of OOP topics by writing reports	TU
G3.1	Apply OO concepts to analyze problems	TU
G3.2	Apply OO concepts to design solution architectures for problems	TU

G4.1	Explain and discuss different OO concepts and principles: class, object, data hiding, encapsulation, inheritance, dynamic binding...	T
G5.1	Distinguish the differences between C++, C#, and Java regarding to some OOP features	T
G6.1	Use UML class diagram to illustrate the OO architecture of problems	TU
G6.2	Use C++ to solve problems using object-oriented programming	TU
G6.3	Apply the header and implementation files separation in terms of program design when solving object-oriented problems.	TU

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Introduction to Object-Orient Concepts and Programming	G1.1, G2.1	Lecturing Q&A, Group discussion (Chapter 1)
2	Class, constructors and destructor	G1.1, G2.1, G2.2, G3.1, G4, G5	Lecturing Demonstration, Q&A (Chapter 3, chapter 10 of Text book2)
3	Operator overloading	G1.2, G4, G5	Lecturing Group work and discussion HW1: Constructors & operators (Chapter 3, chapter 11 of

			Text book2)
4	Inheritance	G1.2, G3.1, G4	Lecturing Demonstration, discussion HW2: Operator & Inheritance A21: Project given (Chapter 4, chapter 13 of Text book2)
5	Inheritance and polymorphism	G1.1, G3.1, G3.2, G4, G5	Lecturing Demonstration Case study & discussion HW3: Inheritance A22: Topic given (Chapter 6)
6	Const correctness & Review Midterm		Question & answer Case study and discussion HW4: Inheritance A32: Midterm Exam
7	Templates and meta-programming	G1.1, G2.2, G4.1	Lecturing Demonstration HW5: Templates (Chapter 8)
8	Exception handling & RAI	G1.1, G4	Lecturing Q&A, discussion HW6: Exception handling & OOP (Chapter 10, chapter 15 of Textbook2)

9	Standard template library (STL): vector, string, streams	G1.1, G2.2, G4.1	Lecturing HW7: STL (Chapter 9, chapter 16 of Text book2)
10	Design patterns: Singleton, Composite Review for the final exam	G1.2, G2.2, G3.1, G3.2, G4	Case study, discussion Demonstration (GoF Book) Q&A, Discussion Project submitted

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			10%
A11	LabWork & Homework: HW1, HW2, ...HW7	HW1 to HW7: practicing and implementing based on knowledge taught in class A report on the comparison between C++ and C#/Java	G1.1, G2, G3, G4, G5, G6.2, G6.3	10%
A2	Project & Seminar			15%

A21	Project	Perform the analysis, design and problem solving skills based on OOP on a real-world problem	G1.2, G2.1, G2.2, G3, G4, G6	7.5%
A22	Seminar	Group work on design patterns: report & presentation	G1.2, G2.2, G3, G4, G6	7.5%
A3	Exams			75%
A31	Lab final exam	Lab exam on computer	G1.1, G3.1, G6.1	10%
A32	Midterm exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems	G1.1, G3, G4, G6.1, G6.2	25%
A33	Final exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems	G1.1, G3, G4, G6.1, G6.2	40%

7. RESOURCES

Textbooks

- *C++ Primer Plus*, Stephen Prata, 6th edition, Addison-Wesley, 2011.

Others

- The C++ Programming Language, Bjarne Stroustrup.
- *Design Patterns: Elements of Reusable Object-Oriented Software*, Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides, Addison-Wesley, 1995.
- *Effective Modern C++: 42 Specific Ways to Improve Your Use of C++11 and C++14 1st Edition*, Scott Meyers, O'Reilly Media, Incorporated, 1st edition, 2014.
- *Phương pháp lập trình hướng đối tượng*, Trần Đan Thư, Đinh Bá Tiến, Nguyễn Tấn Trần Minh Khang, Nhà xuất bản Khoa học và Kỹ thuật, 2018.

IDE

Visual Studio C++

Dev C/C++

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.

Dean of the Faculty



Dinh Ba Tien